

Automated Interpretable Neuron Screening for Calcium Imaging

Nikita Pospelov

*Laboratory of Neuronal Intelligence
Institute for Advanced Brain Studies
Lomonosov Moscow State University
Moscow, Russia
pospelov.na14@physics.msu.ru*

Olga Rogozhnikova

*Laboratory of Neuronal Intelligence
Institute for Advanced Brain Studies
Lomonosov Moscow State University
Moscow, Russia
osrogozhnikova@gmail.com*

Viktor Plyusnin

*Laboratory of Neuronal Intelligence
Institute for Advanced Brain Studies
Lomonosov Moscow State University
Moscow, Russia
witkax@mail.ru*

Anna Ivanova

*Laboratory of Neuronal Intelligence
Institute for Advanced Brain Studies
Lomonosov Moscow State University
Moscow, Russia
anivis33@gmail.com*

Ksenia Toropova

*Laboratory of Neuronal Intelligence
Institute for Advanced Brain Studies
Lomonosov Moscow State University
Moscow, Russia
xen.alexander@gmail.com*

Olga Ivashkina

*Laboratory of Neuronal Intelligence
Institute for Advanced Brain Studies
Lomonosov Moscow State University
Moscow, Russia
oivashkina@gmail.com*

Konstantin Anokhin

*Laboratory of Neuronal Intelligence
Institute for Advanced Brain Studies
Lomonosov Moscow State University
Moscow, Russia
k.anokhin@gmail.com*

Abstract—Manual curation of candidate neurons is a persistent bottleneck in one-photon calcium imaging: component-extraction algorithms deliberately over-segment to preserve sensitivity, leaving experts to inspect hundreds to thousands of candidates per session by hand. We present an automated, fully transparent screening system. First, redundant components are removed by a graph-based procedure that treats candidates as nodes of a temporal-correlation network and collapses its densely connected clusters. Each surviving candidate is then described by 36 interpretable features spanning spatial morphology, trace statistics, wavelet-derived calcium-event properties, and reconstruction quality. Calcium transients are detected in the time-scale plane via a continuous wavelet transform with generalized Morse wavelets, and a five-tier cascade restores event features for neurons on which detection fails, lifting coverage from 42% to 80%. Classification uses an Explainable Boosting Machine optimized for an asymmetric F_β objective ($\beta = 0.577$) that penalizes false positives three times more than false negatives. Under session-stratified cross-validation on 92,242 neurons from 187 sessions across four paradigms, the system attains 0.976 precision and 0.933 recall (AUC 0.978), cutting curation from hours to minutes with a per-feature audit trail for every decision.

Index Terms—calcium imaging, quality control, correlation networks, wavelet event detection, interpretable machine learning, explainable boosting machine

This work was supported by the Non-commercial Foundation for the Advancement of Science and Education “INTELLECT”. N.P. acknowledges the support of the Brain program of the IDEAS Research Center.

I. INTRODUCTION

One-photon calcium imaging with miniaturized microscopes (miniscopes) records the activity of large neuronal populations in freely behaving animals. Single-neuron signals are extracted by constrained matrix-factorization algorithms such as CNMF-E [1], implemented in the CaImAn suite [2]. By design these algorithms favor sensitivity over specificity: they over-segment the field of view so that no genuine neuron is missed, accepting that many false candidates—from motion, neuropil contamination, imaging aberrations, edge effects, and over-splitting of single cells into several correlated components—are produced alongside real neurons.

The standard workflow therefore ends with a manual step in which an expert inspects every candidate, weighing its spatial footprint, temporal activity, and quality metrics, and labels it a neuron or an artifact. For a typical session of 500–1500 candidates this requires one to three hours, and the implicit criteria that distinguish a “real” neuron are inconsistent between annotators and hard to transfer.

We automate this bottleneck with an interpretable pipeline whose contribution is fourfold: (i) a graph-based duplicate-removal stage that formulates redundancy elimination as clustering of a temporal-correlation network; (ii) a 36-dimensional interpretable feature space, including calcium-event descriptors extracted by continuous-wavelet analysis; (iii) a glass-box Explainable Boosting Machine (EBM) trained against

an asymmetric-cost objective; and (iv) validation at scale—92,242 expert-labeled neurons from 187 sessions—yielding expert-level accuracy with a complete decision audit trail.

II. METHODS

A. Animals, imaging, and preprocessing

Experiments were performed on male and female C57Bl/6 mice aged 3–5 months. Activity of CA1 neurons was recorded with a miniature microscope (UCLA Miniscope V4.4, OpenEphys). Animals underwent stereotaxic surgery in which viral particles carrying the gene of the calcium sensor GCaMP6s were injected into hippocampal area CA1, a GRIN lens (1 mm) was implanted, and a miniscope baseplate was fixed. Behavior was captured with a video camera (FLIR Chameleon 3) and synchronized with the imaging stream in Bonsai [3]. All procedures were approved by the Animal Care Committee of Lomonosov Moscow State University (application No. 159-a, Bioethics Commission meeting No. 154-d of 17.08.2023) and complied with Order No. 267 of the Ministry of Health of the Russian Federation and the US NIH Guide for the Care and Use of Laboratory Animals. Video was motion-corrected with NoRMCorre [4] and decomposed with CNMF-E [1] in the authors' BEARMIND package; traces were normalized as dF/F and footprints registered across sessions with CellReg [5]. The resulting candidates form a labeled dataset of **92,242 neurons from 187 sessions** across four paradigms (LNOF: 49,767; NOF: 30,398; RFC: 7,505; FOF: 4,572), each manually labeled as a genuine neuron (81.1%) or an artifact (18.9%).

B. Network-based duplicate removal

CNMF-E frequently splits one physical neuron into several highly correlated components. We eliminate this redundancy by clustering a temporal-correlation network of candidates. Let $\rho_{ij}^{(s)}$ be the Spearman correlation between the dF/F traces of components i, j in temporal block s . Each block is binarized, $B_{ij}^{(s)} = \mathbb{I}[\rho_{ij}^{(s)} \geq \rho_0]$ with $\rho_0 = 0.4$, and co-activation is accumulated over S blocks into a weighted adjacency

$$A_{ij} = \left(\sum_{s=1}^S B_{ij}^{(s)} \right) \mathbb{I} \left[\sum_s B_{ij}^{(s)} \geq m_0 \right], \quad A_{ii} = 0, \quad (1)$$

where $m_0 = 3$ requires reproducible co-activation. On the graph $G = (V, A)$ whose nodes are candidate components, the connected components $\{C_k\}$ (found by breadth-first search) with $|C_k| > 1$ are duplicate groups. Each group is ranked by its internal correlation density

$$d(C) = \frac{1}{|C|(|C| - 1)} \sum_{i \neq j \in C} \bar{\rho}_{ij}, \quad (2)$$

and groups are resolved in decreasing d . Within a group, two components whose footprints lie within $\delta = 5$ px (boundary-to-boundary) are taken to be the same neuron: the higher-SNR copy is kept, and the other is merged when $|\Delta \text{SNR}| \leq \tau$ ($\tau = 10$) or discarded otherwise. This collapses dense subgraphs to single representatives before classification.

C. Interpretable feature space

Each surviving candidate is described by 36 features in six groups: *spatial/morphological* (11; area, circularity, convexity, aspect ratio, eccentricity, edge distance, local density, ...), *temporal/trace* (9; skewness, kurtosis, baseline drift, noise, bimodality, Hurst exponent, ...), *event-based* (8; rate, fit quality, signal-to-noise, rise/decay times, amplitude variability, half-crossing rate), *reconstruction* (4; R^2 , NMAE, NRMSE, residual SNR of a calcium-kinetics fit), *extractor-provided* (2; CalmAn SNR and spatial-correlation score), and *kinetics-provenance* (2; see Sec. II-E). No single metric separates neurons from artifacts; the discriminative power is distributed across these complementary views.

D. Wavelet-based event characterization

The most discriminative features derive from calcium transients, which we detect in the time-scale plane. After min-max normalization and Gaussian smoothing, the trace $x(t)$ is mapped by the continuous wavelet transform

$$W(a, b) = \frac{1}{\sqrt{a}} \int x(t) \psi^* \left(\frac{t - b}{a} \right) dt, \quad (3)$$

with a generalized Morse wavelet [6], whose frequency-domain form is $\Psi_{\beta, \gamma}(\omega) = \mathbb{I}_{\omega > 0} \omega^\beta e^{-\omega^\gamma}$, $(\beta, \gamma) = (2, 3)$, as implemented in the DRIADA toolkit [7]. This choice gives a nearly symmetric, low-oscillation analytic wavelet well matched to the asymmetric rise/decay of calcium events; scales a are log-spaced to span transient durations 0.275–2.25 s. Local maxima of $|W(a, \cdot)|$ are linked across scales into ridges, and a ridge is accepted as an event when it persists over $\geq L$ scales, reaches a maximum-magnitude scale index $\geq \sigma^*$, peak magnitude $\geq a_0$, and duration $\leq \Delta$. From the event support $E \subset \{1, \dots, T\}$ we form, e.g., the event signal-to-noise ratio

$$\text{SNR}_{\text{ev}} = \log \left(1 + \frac{\text{med}_k A_k - \text{med}(x_{\bar{E}})}{\text{MAD}(x_{\bar{E}})} \right), \quad (4)$$

where A_k are per-event peak amplitudes and $x_{\bar{E}}$ the inter-event baseline, and the event-restricted reconstruction quality

$$R_{\text{ev}}^2 = 1 - \frac{\sum_{t \in E} (x_t - \hat{x}_t)^2}{\sum_{t \in E} (x_t - \bar{x}_E)^2}, \quad (5)$$

against a single-compartment kinetics fit $\hat{x}$.

E. Cascading kinetics estimation

Ridge-based detection fails on 40–60% of candidates (low SNR, sparse firing, nonstationary baseline), leaving eight event features undefined. A five-tier cascade restores them: (i) wavelet detection with strict criteria; (ii) wavelet with relaxed thresholds (≥ 3 events, $R_{\text{ev}}^2 \geq 0.6$); (iii)–(iv) a MAD-threshold detector ($n_{\text{MAD}} = 4$) at strict then relaxed criteria; and (v) physically plausible defaults ($t_{\text{rise}} = 0.25$ s, $t_{\text{off}} = 2.0$ s). The first successful tier is recorded as a categorical *kinetics-provenance* feature, letting the classifier discount low-reliability event evidence. Coverage rises from 42% (single attempt) to 80%, and spatial, trace, and reconstruction features remain defined even at tier (v)—graceful degradation rather than outright failure.

F. Classifier and decision objective

Candidates are classified by an Explainable Boosting Machine (EBM) [8], [9], a generalized additive model

$$g(\mathbb{E}[y | \mathbf{x}]) = \beta_0 + \sum_j f_j(x_j) + \sum_{(j,k) \in \mathcal{I}} f_{jk}(x_j, x_k), \quad (6)$$

with per-feature shape functions f_j and $|\mathcal{I}| = 20$ learned pairwise interactions. Unlike black-box ensembles, every prediction decomposes exactly into additive feature contributions, yielding a complete audit trail while matching the accuracy of random forests and gradient boosting. Because errors are asymmetric—a false positive (artifact kept) can corrupt population statistics, whereas a false negative merely costs statistical power—we optimize

$$F_\beta = (1 + \beta^2) \frac{\text{precision} \cdot \text{recall}}{\beta^2 \text{precision} + \text{recall}}, \quad (7)$$

with $\beta = \sqrt{1/3} \approx 0.577$ [10], weighting false positives three times more than false negatives. Hyperparameters were chosen by stratified cross-validation over 128 configurations. At deployment, candidates may be filtered by transparent rules, by EBM score at a tunable threshold, or by a hybrid that applies rules first and the EBM to the remainder.

III. RESULTS

A. Classification performance

We evaluate by cross-validation stratified by *recording session*, so that neurons from one session never appear in both training and test folds; this precludes exploiting session-specific imaging conditions. Across folds (mean $\pm$ s.d.) the production model (v9) attains $F_\beta = 0.965 \pm 0.004$, precision 0.976 ± 0.003 , recall 0.933 ± 0.007 , and AUC 0.978 ± 0.005 . A precision of 97.6% means only 2.4% of accepted candidates are artifacts; recall of 93.3% means 6.7% of genuine neurons are missed—a trade-off deliberately favoring precision. The low cross-session variance confirms stable generalization across paradigms.

B. Which features matter

Feature-importance analysis (Fig. 1) shows that temporal trace statistics carry the strongest discriminative power: trace skewness and kurtosis rank first and second, validating the heavy-tailed, “spiky” signature of genuine calcium dynamics. Wavelet-derived event descriptors (Eqs. 4–5) rank immediately below, whereas the extractor’s own amplitude SNR ranks lowest—temporal structure is far more informative than raw amplitude.

C. Tuning the operating point

Because the EBM emits a calibrated score, a single threshold tunes the precision/recall balance (Fig. 2). The unconstrained F_β optimum lies at a threshold of 0.66; the deployed operating point (0.75) sits above it, trading a little recall for additional precision, consistent with the asymmetric-cost objective. Lower thresholds (0.3–0.4) maximize recall for exploratory screening, higher thresholds for publication-grade datasets.

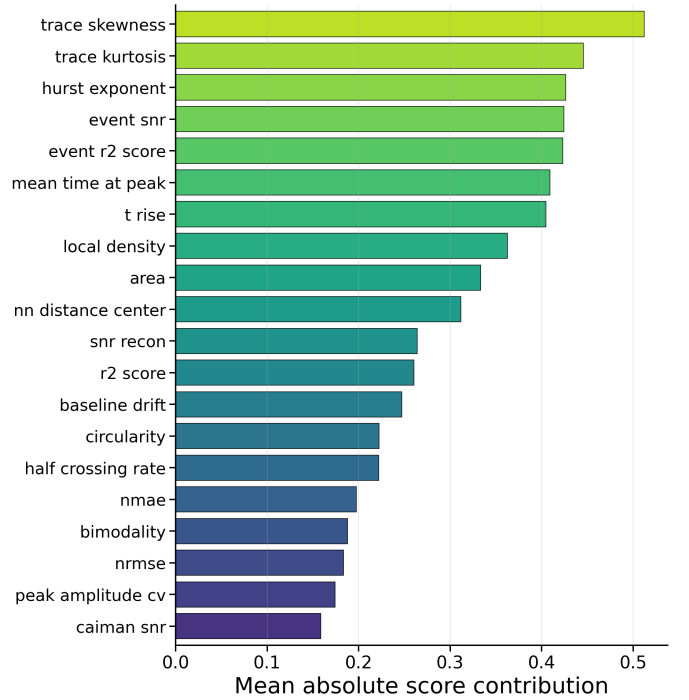


Fig. 1. Mean absolute score contribution of the top features in the production EBM. Temporal trace statistics (skewness, kurtosis) and the Hurst exponent dominate, followed by wavelet event descriptors (event SNR, event R^2); the extractor-provided CalmAn SNR ranks lowest.

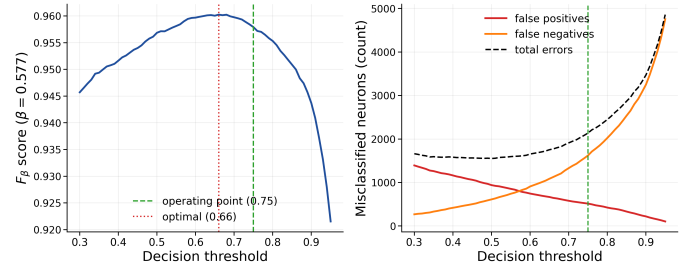


Fig. 2. Threshold analysis on the held-out test split. Left: F_β ($\beta=0.577$) versus decision threshold, with the deployed operating point (0.75) and the F_β optimum (0.66) marked. Right: false positives, false negatives, and total errors versus threshold.

D. Practical impact

Manual curation of a 1000-candidate session takes one to two hours; duplicate removal, feature extraction, and classification together process the same session in two to three minutes—a 30–40 $\times$ speedup enabling same-day analysis. Identical criteria applied to every candidate make the output reproducible, and the additive decomposition of Eq. 6 shows exactly why each decision was made.

IV. DISCUSSION AND CONCLUSION

We have presented an automated quality-control system that closes the last manual step in calcium-imaging preprocessing. Casting duplicate removal as clustering of a temporal-correlation network (Sec. II-B) links a practical engineering

need to the analysis of correlation graphs, and the wavelet front end (Sec. II-D) supplies the features that prove most discriminative. Interpretability is functional, not cosmetic: the additive structure of Eq. 6 exposes which features drove a decision, so a corrected label becomes a training example for the next iteration. Across eight such iterations the model gained measurably in precision and AUC on initially hard cases.

Limitations remain: the system was trained on one-photon GCaMP recordings of hippocampal and cortical neurons, so transfer to other modalities or cell types may require retraining; entirely silent neurons are inherently ambiguous; and $\sim 20\%$ of candidates still rely on default kinetics. Natural extensions include tighter coupling to upstream extractors, real-time assessment during acquisition, and session-specific calibration. More broadly, the work shows that interpretable machine learning can match expert judgment in a scientific decision task while remaining auditable, scaling human expertise rather than hiding it behind opaque automation; combined with CNMF-E extraction it enables a fully automated pipeline from raw miniscope video to verified neural populations.

REFERENCES

- [1] P. Zhou, S. L. Resendez, J. Rodriguez-Romaguera, J. C. Jimenez, S. Q. Neufeld, A. Giovannucci, J. Friedrich, E. A. Pnevmatikakis, G. D. Stuber, R. Hen, M. A. Kheirbek, B. L. Sabatini, R. E. Kass, and L. Paninski, "Efficient and accurate extraction of in vivo calcium signals from microendoscopic video data," *eLife*, vol. 7, p. e28728, Feb. 2018. [Online]. Available: <http://dx.doi.org/10.7554/eLife.28728>
- [2] A. Giovannucci, J. Friedrich, P. Gunn, J. Kalfon, B. L. Brown, S. A. Koay, J. Taxidis, F. Najafi, J. L. Gauthier, P. Zhou, B. S. Khakh, D. W. Tank, D. B. Chklovskii, and E. A. Pnevmatikakis, "CaImAn: An open source tool for scalable calcium imaging data analysis," *eLife*, vol. 8, p. e38173, Jan. 2019. [Online]. Available: <http://dx.doi.org/10.7554/eLife.38173>
- [3] G. Lopes, N. Bonacchi, J. Frazão, J. P. Neto, B. V. Atallah, S. Soares, L. Moreira, S. Matias, P. M. Itskov, P. A. Correia *et al.*, "Bonsai: an event-based framework for processing and controlling data streams," *Frontiers in Neuroinformatics*, vol. 9, p. 7, 2015.
- [4] E. A. Pnevmatikakis and A. Giovannucci, "NoRMCorre: An online algorithm for piecewise rigid motion correction of calcium imaging data," *Journal of Neuroscience Methods*, vol. 291, pp. 83–94, Nov. 2017. [Online]. Available: <http://dx.doi.org/10.1016/j.jneumeth.2017.07.031>
- [5] L. Sheintuch, A. Rubin, N. Brande-Eilat, N. Geva, N. Sadeh, O. Pinchasof, and Y. Ziv, "Tracking the same neurons across multiple days in Ca²⁺ imaging data," *Cell Reports*, vol. 21, no. 4, pp. 1102–1115, Oct. 2017. [Online]. Available: <http://dx.doi.org/10.1016/j.celrep.2017.10.013>
- [6] S. C. Olhede and A. T. Walden, "Generalized morse wavelets," *IEEE Transactions on Signal Processing*, vol. 50, no. 11, pp. 2661–2670, 2002.
- [7] Y. Sych, M. Chernysheva, L. T. Sumanovski, and F. Helmchen, "High-density multi-fiber photometry for studying large-scale brain circuit dynamics," *Nature Methods*, vol. 16, no. 6, pp. 553–560, May 2019. [Online]. Available: <http://dx.doi.org/10.1038/s41592-019-0617-2>
- [8] Y. Lou, R. Caruana, J. Gehrke, and G. Hooker, "Accurate intelligible models with pairwise interactions," in *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 2013, pp. 623–631.
- [9] H. Nori, S. Jenkins, P. Koch, and R. Caruana, "InterpretML: A unified framework for machine learning interpretability," *arXiv preprint arXiv:1909.09223*, 2019. [Online]. Available: <https://arxiv.org/abs/1909.09223>
- [10] C. J. van Rijsbergen, *Information Retrieval*, 2nd ed. Newton, MA, USA: Butterworth-Heinemann, 1979.